

MTH209: Worksheet 4

Random Vectors and Covariance Matrices

In many statistical problems, we observe not a single random variable, but a *collection* of random variables measured simultaneously.

Examples include:

- height, weight, and blood pressure of a patient
- returns of multiple stocks on the same day
- coordinates of a random point in space
- repeated measurements taken at the same time

To study such data rigorously, we introduce the concept of **random vectors** and their associated **covariance structure**.

These ideas form the foundation for multivariate statistics, principal component analysis, multivariate normal models, and modern machine learning methods.

Random Vectors

A **random vector** is a collection of random variables considered jointly. Let

$$X = (X_1, X_2, \dots, X_d)^\top$$

be a d -dimensional random vector, where each X_i is a real-valued random variable defined on the same probability space. The joint behavior of the components is described through the **joint distribution**

$$F_X(x_1, \dots, x_d) = \mathbb{P}(X_1 \leq x_1, \dots, X_d \leq x_d).$$

Unlike univariate random variables, random vectors can exhibit **dependence** across components, and this is why they are of critical importance.

The **mean vector** of X is

$$\mu = \mathbb{E}[X] = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_d])^\top.$$

Covariance Matrix

While the mean vector captures location, it does not describe how the components vary or interact. This role is played by the **covariance matrix**. The **covariance matrix** of a random vector X is defined as

$$\Sigma = \text{Cov}(X) = \mathbb{E}[(X - \mu)(X - \mu)^T].$$

Its (i, j) -th entry is

$$\Sigma_{ij} = \text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])].$$

Key properties:

- Σ is symmetric: $\Sigma_{ij} = \Sigma_{ji}$
- Σ is positive semidefinite
- Diagonal entries are variances: $\Sigma_{ii} = \text{Var}(X_i)$
- Σ_{ij} measures *linear association* between X_i and X_j

Geometrically, the covariance matrix describes the **spread and orientation** of the distribution. Positive covariance indicates that large values of one variable tend to occur with large values of the other, while negative covariance indicates opposite movement.

Effect of Translation and Rotation

Let $Y = X + a$ where this addition is element-wise and $a \in \mathbb{R}^d$ is a constant vector. Then

$$\mathbb{E}[Y] = \mu + a,$$

but $\text{Cov}(Y) = \text{Cov}(X)$. Thus, translation changes the mean but not the covariance.

On the other hand, let $Y = AX$, where A is a $d \times d$ matrix. Then

$$\text{Cov}(Y) = A\Sigma A^\top.$$

If A is an orthogonal matrix (i.e., $A^\top A = I$), then A represents a **rotation or reflection**. Rotations do not change the total variability, but they redistribute variance across coordinates. This is essentially what we did in PCA last week!

```
set.seed(1)

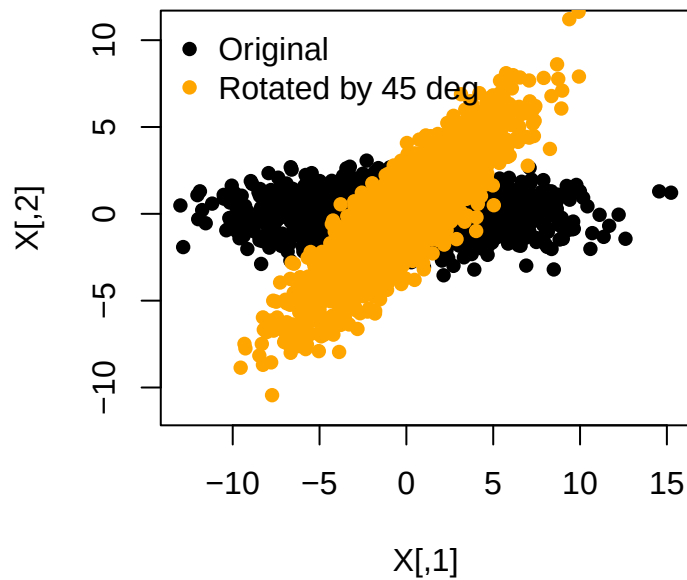
n <- 2000
X <- matrix(c(rnorm(n, sd = 4), rnorm(n, sd = 1)), ncol = 2)

theta <- pi/4
R <- matrix(c(cos(theta), -sin(theta),
              sin(theta),  cos(theta)), 2, 2, byrow = TRUE)

Y <- X %*% t(R)

plot(X, asp = 1, main = "Data cloud", pch = 16)
points(Y, col = "orange", pch = 16)
legend("topleft", legend = c("Original", "Rotated by 45 deg"),
      col = c("black", "orange"), pch = 16, bty = "n")
```

Data cloud



Covariance and Independence

If

$$\Sigma = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix},$$

then $\text{Cov}(X_1, X_2) = 0$. This means X_1 and X_2 are **uncorrelated**, but not necessarily independent. Independence requires that for all sets A, B

$$\mathbb{P}(X_1 \in A, X_2 \in B) = \mathbb{P}(X_1 \in A) \mathbb{P}(X_2 \in B)$$

There exist random variables with zero covariance but strong nonlinear dependence.

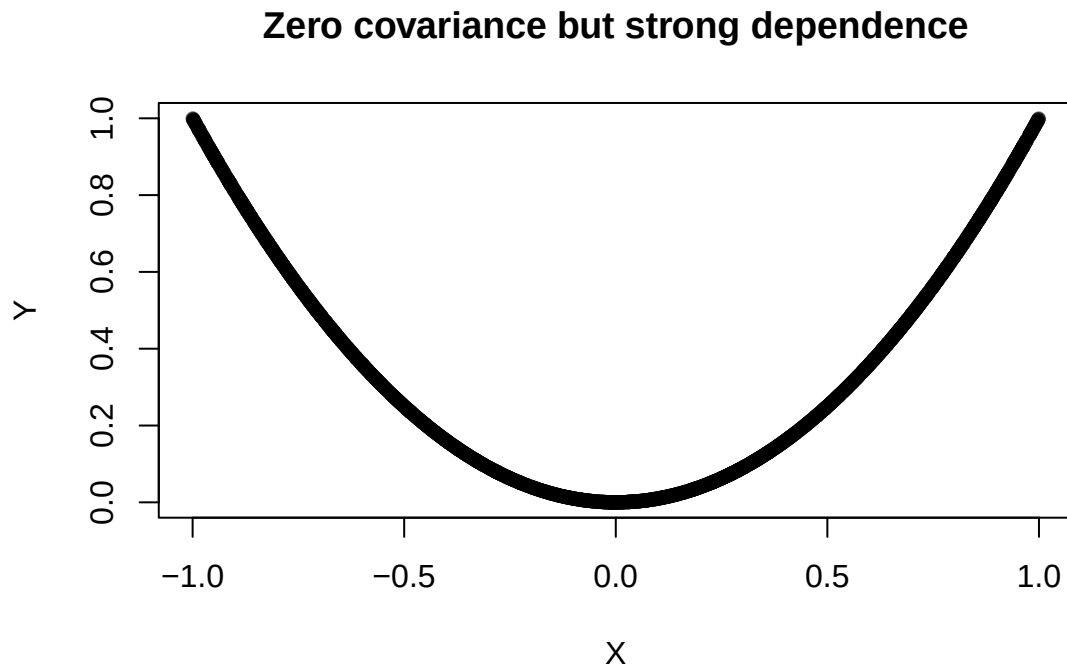
```
set.seed(2)

n <- 5000
X <- runif(n, -1, 1)
Y <- X^2

cov(X, Y)
```

```
[1] 0.001847299
```

```
plot(X, Y, pch=16, col=rgb(0,0,0,0.25),
     main="Zero covariance but strong dependence")
```



Multivariate Normal Distribution

A random vector X that follows a **multivariate normal** distribution with mean μ and covariance Σ has density

$$f(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right).$$

Key properties:

- Linear transformations of a normal vector remain normal
- Marginal distributions are normal
- Conditional distributions are normal

Most importantly:

For multivariate normal random vectors, zero covariance **does imply independence**.

This is a special and unique property of the normal distribution. Thus, when X is jointly normal,

$$\Sigma \text{ diagonal} \Rightarrow X_1, \dots, X_d \text{ independent.}$$

Correlation Matrix

Covariance depends on scale. To remove scale effects, we use **correlation**.

The correlation between X_i and X_j is

$$\rho_{ij} = \frac{\text{Cov}(X_i, X_j)}{\sqrt{\text{Var}(X_i)\text{Var}(X_j)}}.$$

The **correlation matrix** is

$$R = D^{-1/2}\Sigma D^{-1/2},$$

where D is the diagonal matrix of variances.

Properties:

- Diagonal entries are 1
- All entries lie between -1 and 1
- Dimensionless and scale-free

The above are also reasons why it is often recommended to do PCA on the correlation matrix (scale the columns to variance 1) than the covariance matrix.

Correlation measures **linear association**, not general dependence.

Estimating Covariance and Correlation Matrices

Suppose we observe independent realizations

$$X^{(1)}, \dots, X^{(n)} \in \mathbb{R}^d.$$

Naturally the estimator of the mean vector μ is:

$$\bar{X} = \frac{1}{n} \sum_{k=1}^n X^{(k)}.$$

As discussed in the previous week, an estimator of the covariance matrix Σ is

$$\hat{\Sigma} = \frac{1}{n-1} \sum_{k=1}^n (X^{(k)} - \bar{X})(X^{(k)} - \bar{X})^T.$$

The sample correlation matrix is obtained by standardizing each variable using its sample variance and then computing covariances.

Both covariance and correlation matrices play a central role in:

- Multivariate modeling
- Principal component analysis
- Multivariate normal theory
- Dependence modeling

Questions

1. (Homework) Is the sample covariance matrix guaranteed to be non-singular?
2. Consider a bivariate normal random vector

$$X \sim \mathcal{N}(\mu, \Sigma)$$

with

$$\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}.$$

- a. Generate $n = 5000$ samples from this distribution using `rmnorm` and matrix operations only.
- b. Create a scatter plot of the generated data.
- c. Estimate the sample covariance matrix.
- d. Estimate the sample correlation matrix.
- e. Repeat the analysis with

$$\Sigma = \begin{pmatrix} 5 & -1 \\ -1 & 1 \end{pmatrix}$$

and compare the geometry of the two point clouds.

3. Let $\Theta \sim \text{Uniform}(0, 2\pi)$ and define the random vector

$$X = \begin{pmatrix} \cos(\Theta) \\ \sin(\Theta) \end{pmatrix}.$$

- a. Generate $n = 5000$ samples of Θ .
 - b. Construct $(X_1, X_2)^\top = (\cos \Theta, \sin \Theta)^\top$.
 - c. Create a scatter plot of the data.
 - d. Estimate the covariance matrix.
 - e. Estimate the correlation coefficient.
4. What is the computational complexity of a sample covariance matrix calculated from an $n \times p$ matrix of data?
 5. What is the computational complexity of a sample correlation matrix calculated from an $n \times p$ matrix of data?
 6. Using `system.time()` compute sample covariance matrices for a large $n \times p$ matrix. Use your imagination for what “large” means.

7. The dataset `EuStockMarkets` is available in base R and contains daily closing prices of four major European stock indices:

- DAX (Germany)
- SMI (Switzerland)
- CAC (France)
- FTSE (United Kingdom)

a. Calculate the sample covariance and sample correlation matrices.

b. Do you think there is anything wrong with the calculation in part a.?

c. Calculate the log returns for each stock: $\log(X_t) - \log(X_{t-1})$

d. Calculate the sample correlation matrix now for these log returns. Is there any difference from before?

e. Use the following code to make a heatmap:

```
corrplot(  
  cor_matrix,  
  method = "color",  
  type = "upper",  
  addCoef.col = "black",  
  tl.col = "black",  
  tl.srt = 45  
)
```

8. A dataset wine is provided to you in a txt file. More information is available here: <https://archive.ics.uci.edu/dataset/109/wine>

a. Load this data and calculate the sample covariance and sample correlation matrix (removing the first column of the kind of wine)

b. Implement PCA. What do you observe? Can we fix this?

c. Implement PCA on the correlation matrix now. Now what do you observe?

d. Plot the first two PC score components, and color the points based on the Class of wine. What do you observe?

e. What would be the choice of k ?